Exploring the Relationship Between Sunspot Activity and Wheat Production Using Time-Series Analysis and Moving Averages

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Abstract:

This project examines the impact of sunspots on wheat yields. Sunspots are temporary phenomena on the Sun's surface that increase solar output, thereby affecting Earth's temperature and exposure to solar radiation. Sunspots follow an 11-year solar cycle and may influence food production.

Using RStudio, we analyzed sunspot and wheat yield data and explored the correlations between them. The sunspot data came from the Global Surface Summary of the Day (GSOD), maintained by the National Centers for Environmental Information (NCEI). The wheat yield data came from the U.S. Department of Agriculture (USDA). From 1960 to 2000, the data showed a strong positive correlation, indicating a relationship between higher sunspot activity over the solar cycle (11 years) and higher crop yields (R = 0.619). After 2000, however, the correlation weakened, likely due to advances in agricultural technology, such as precision agriculture, as well as external environmental factors that reduced the relative impact of the solar cycle.

We examined the correlations between sunspots and wheat production, taking into account a delayed effect. We found a stronger correlation between wheat yield and sunspots when using deseasonalized sunspot data from 5 years in the future (R = 0.871). This trend was more consistent compared to the immediate effect of sunspots on wheat yield.

Overall, these findings support the hypothesis that periods of high sunspot activity are associated with higher wheat yields. However, technological advancements may have influenced the wheat yield data and human resilience against the influence of sunspots.

Background Information:

- Sunspots: temporary dark spots on the sun that indicate an increase in solar energy, which means higher temperature, and solar radiation (Fig. 1).
- Temperature and solar radiation impact crop yield.
  - Ex., extra energy, variation in water requirements, solar radiation, and DNA damage.
- Sunspot frequency follows an 11-year cycle (Fig. 2).
- Wheat is one of the most widely cultivated crops worldwide and is consistently monitored. Slight changes in crop yield will be more obvious.
- **Hypothesis:** Because sunspots increase solar energy, sunspots and wheat yield will have a positive correlation.

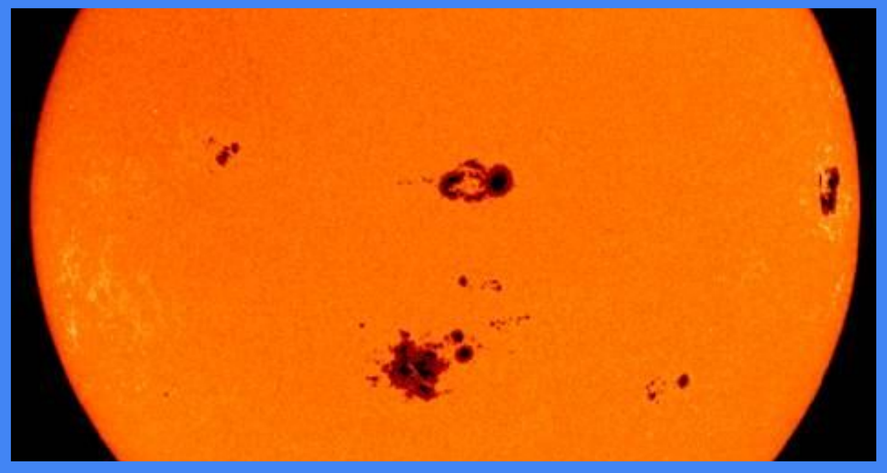

Figure 1: Picture of sunspots on the sun's surface. Adopted from the National Weather Service, 2024.

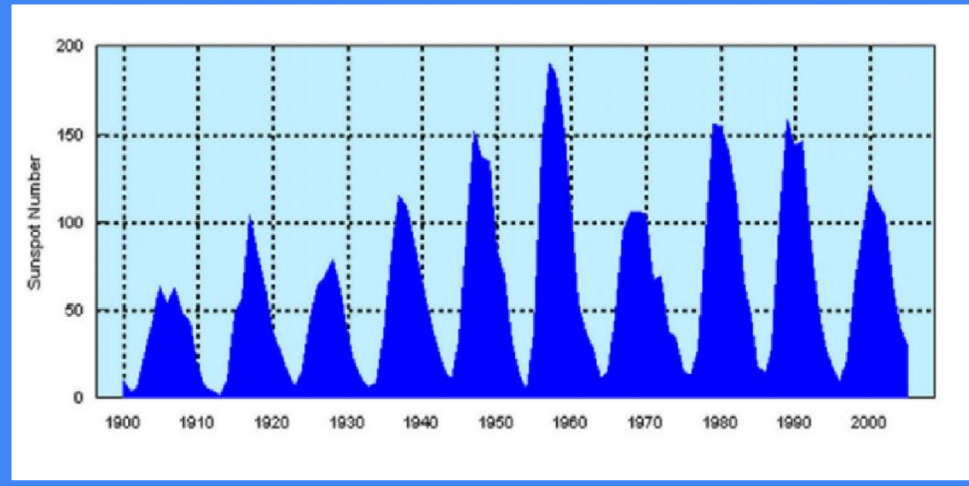

Figure 2: Sunspot frequency from 1900 to 2006. Adapted from NCEI, 2006.

Method and Materials:

- **United States Department of Agriculture (USDA):**
  - Requested a custom query of global wheat yield from 1960 to 2013
- **RStudio (v. 2024.12.1)**
  - Found the annual sunspot frequency from the GSODR data on RStudio.
  - Combined data tables for the year, wheat yield, and sunspots
  - Made a scatterplot of worldwide wheat yield vs. sunspot number (Fig. 3).
  - Type L graph for years vs. sunspot numbers and wheat yield(Fig. 4)
  - Detrended the data with kz moving average: period(m) = 11, k = 3 (Fig. 5)
    - Detrending: removing long-term trends to expose short-term trends.
    - Kz moving avg: replaces a data point with the found average within a timespan(m) around the data point. Repeat on all data k times.
  - Cross correlation function(ccf): found change in wheat yield annually; added kz moving average(m=11, k=3) to sunspots frequency and detrended wheat; plot with max.lag = 20 (Fig. 6)
  - Packages: data.frame, GSODR, tidyverse, kza, lubridate, readxl, writexl

Results:

- No linear correlation is noticeable in the raw data on the scatterplot (Fig. 3) or over the course of years (Fig. 4)
- Detrended wheat yield and deseasonalized sunspot frequency over the course of years show a correlation (Fig. 5)
  - Looking only at data from 1960 to 2000, R=0.619
- When accounting for lag, R=0.871 with a lag of 5 years (Fig. 6)

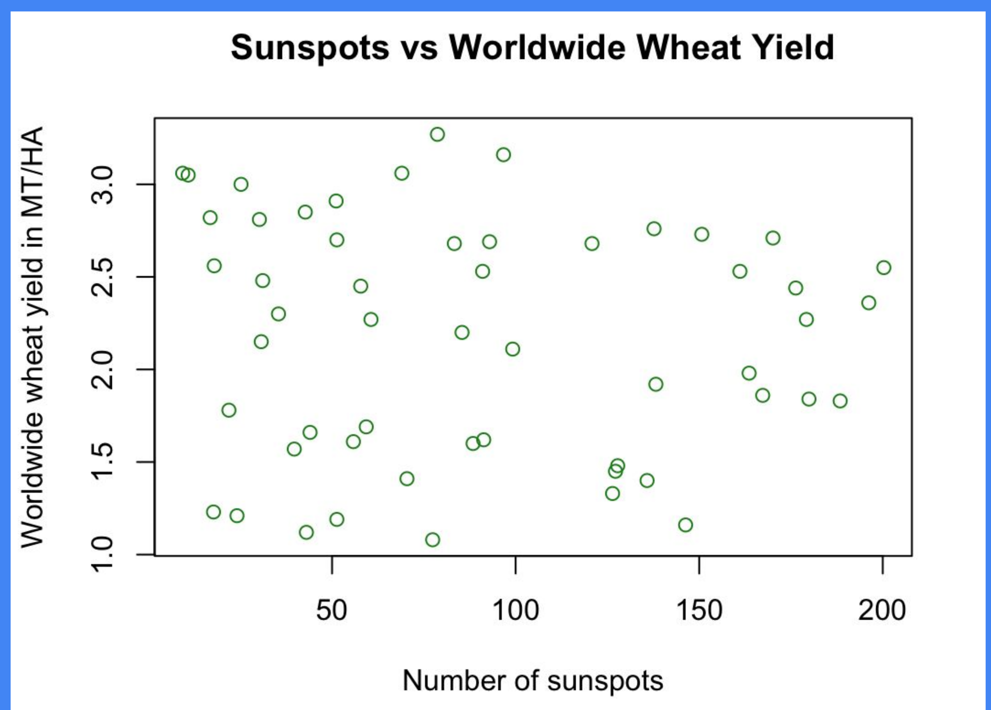

Figure 3: Scatterplot of worldwide wheat yield in a certain year to the sunspot numbers of same year. x-axis = sunspot numbers, y-axis = wheat yield. No obvious correlation

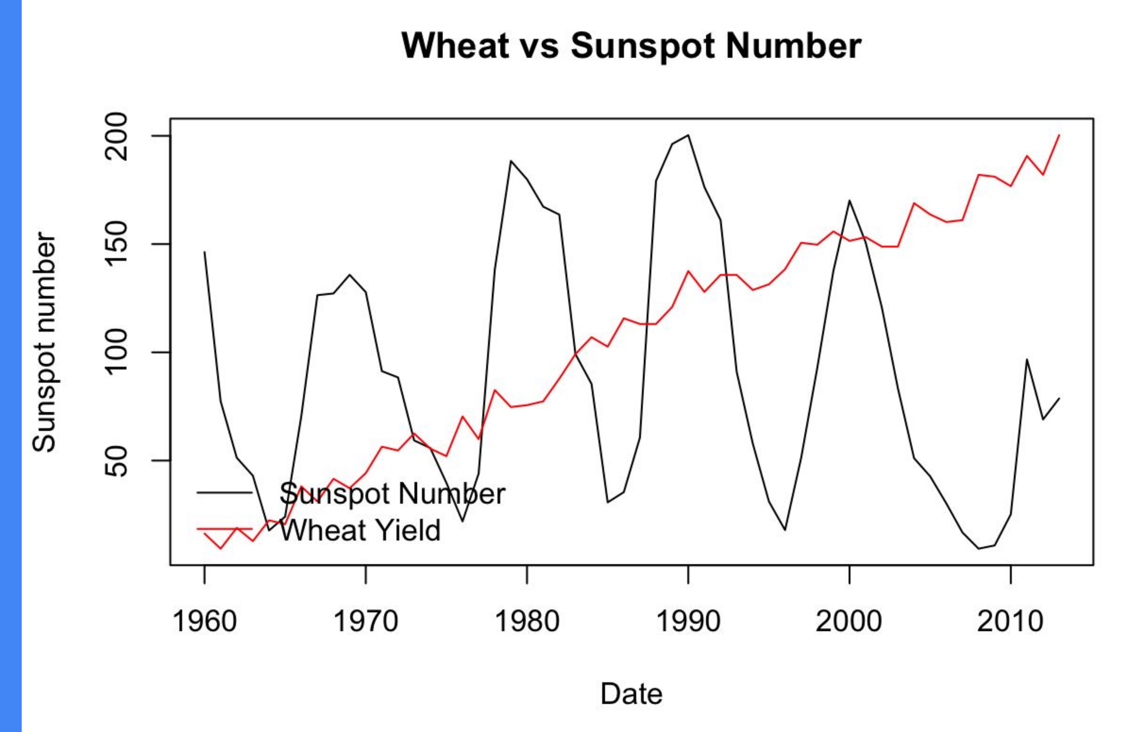

Figure 4: The plot of sunspots to years and wheat yield to years. Solar cycle is visible. No obvious correlation. The wheat yield is steadily increasing, likely due to improvements in agricultural techniques and technology.

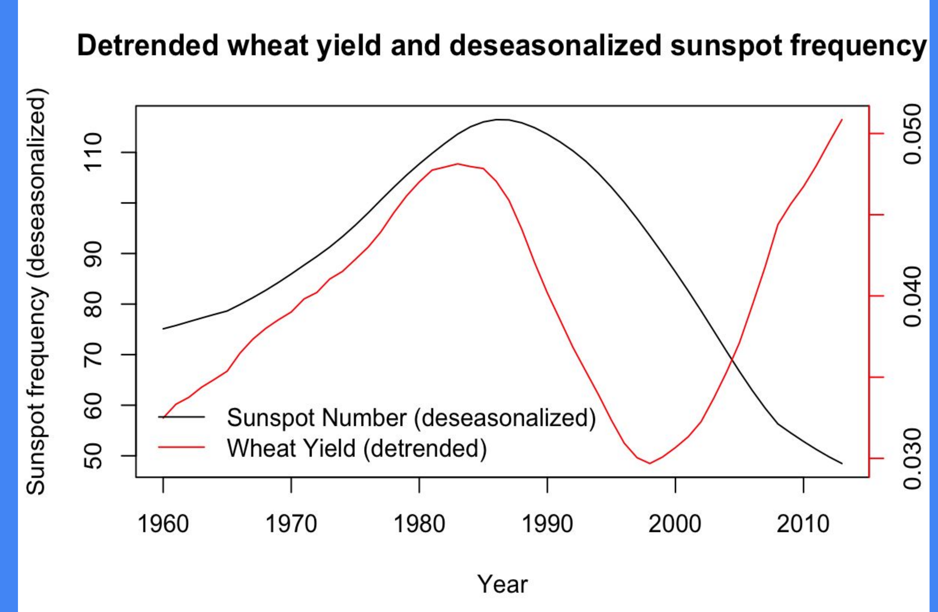

Figure 5: Detrended wheat yield and deseasonalized sunspot frequency per year with a k=3. Wheat data deviates after the year 2000, May reflect advances in agricultural techniques.

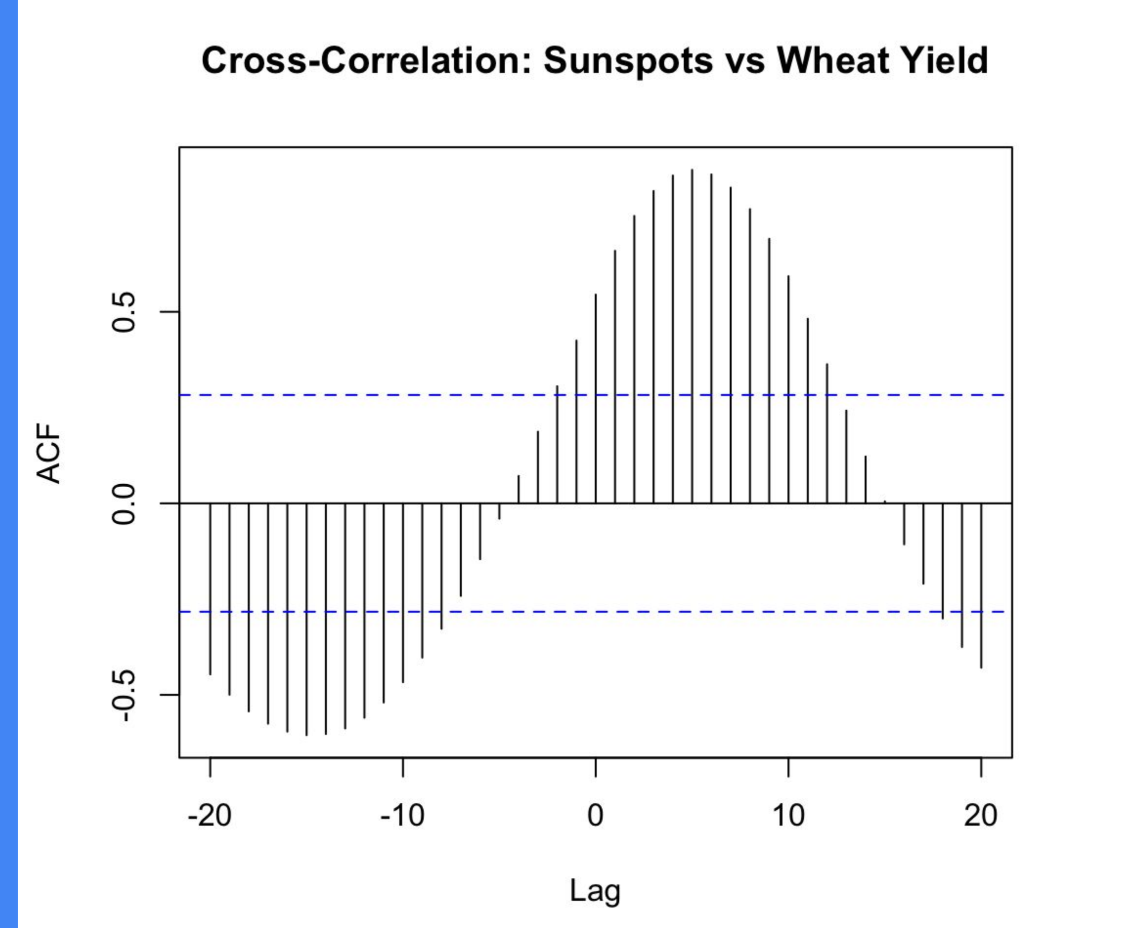

Figure 6: Cross-correlation graph. Compares deseasonalized sunspots to detrended wheat yield up to 20 years earlier (negative lag) or later (positive lag).

Discussion

- Sunspot frequency and wheat yield do correlate.
  - Seen primarily between 1960 and 2000 (Fig. 5)
- Sunspots increase global temperature (~0.1 °C) and ultraviolet (UV) radiation (Scafetta et al, 2014)
- More sunspots mean more solar energy, leading to higher temperatures. So, there can be a longer growing season and earlier sprouting.
  - Although high temperatures mean higher water requirements, higher temperatures can cause more precipitation, which plants can use and absorb
  - Longer and hotter growing seasons can produce more crops.
- UV-A can accelerate photosynthetic rates of wheat sprouts (Joshi et al, 1997)
  - UV-A and UV-B can be detrimental to crops in large quantities (Kataria et al, 2012; Molina-Montenegro et al, 2024)
- The trend disappeared after the year 2000, possibly due to advancements in agriculture.
  - This decoupling could be due to the digital revolution and the rise of precision agriculture

Conclusion

- Sunspots appear to have a positive effect on wheat growth.
  - However, technological and agricultural advancements might have accelerated wheat yields to the point that it is difficult to decipher.
- Limits: quantity of data, data susceptibility to global influences.
- Future research:
  - Research other crop data (ex., soybeans, corn, root vegetables, etc.).
  - Measuring crop growth over the course of years and connecting it to sunspot activity.
  - Narrow the data field to specific locations (e.g., hemisphere, country, continent).
  - How do sunspots influence specific aspects of wheat? Ex. texture, products, price.

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